

Patient Name : MR. DEMO
Age / Gender : 32 Years / Male
Referral Name : SELF

Patient ID : 114
Collection Date : 10/07/2024
Reporting Date : 10/07/2024



HEMATOLOGY			
Complete Blood Count (CBC)			
Test Description	Result	Unit	Reference Range
Hemoglobin (Hb)	10.2	gms/dl	13 - 17
R.B.C Count	5.69	mil./cmm	4.5 - 5.5
Packed Cell Volume (HCT)	339	%	40 - 50
Mean Cell Volume (MCV)	595.78	fL	83 - 101
Mean Cell Haemoglobin (MCH)	17.93	pg	27 - 33
Mean Corpuscular Hb Conc. (MCHC)	3.01	%	32 - 38
RDW-CV	16.3	%	11.60 - 14
Total (WBC) Count	7000	cells/c.mm	4000 - 11000
Differential % WBCs count			
Neutrophil	51	%	45 - 75
Lymphocytes	38	%	20 - 40
Eosinophil	01	%	1 - 6
Monocytes	10	%	2 - 10
Basophils	00	%	0 - 2
Absolute Differential Count			
Absolute Neutrophil Count	3570.00	/c.mm	2000 - 7000
Absolute Lymphocyte Count	2660.00	/c.mm	1000 - 3000
Absolute Monocyte Count	700.00	/c.mm	200 - 1000
Absolute Eosinophils Count, AEC	70.0	/c.mm	20 - 500
Absolute Basophils Count	0.00	/c.mm	20 - 100
Platelet Count	275000	cells/c.mm	150000 - 450000
Mean Platelet Volume (MPV)	6.7	fl	7.2 - 11.7
Platelet Distribution Width (PDW)	15.8	%	10 - 14
PCT (Platelet Crit)	0.18	%	0.17 - 0.37
Peripheral Blood Smear			
RBC Morphology	Normocytic Normochromic		
WBC Morphology	Within Normal Limits		
Platelet Morphology	Adequate on smear		
Advice	Kindly correlate clinically & close follow up.		

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BIOCHEMISTRY

URINE FASTING (BLOOD GLUCOSE LEVEL)

Test Description	Result	Unit	Reference Range
Glucose Fasting	90	mg/dl	70 - 110

Test Information:

A blood sugar test is a procedure that measures the amount of sugar, or glucose, in your blood. Your doctor may order this test to diagnose diabetes. People with diabetes can also use this test to check their diabetes status.

High blood sugar level: (Check your blood sugar levels if you feel any of the following symptoms)

- 1) Increased thirst
- 2) Frequent peeing
- 3) Fatigue (weak, tired feeling)
- 4) Weight loss
- 5) Slow-healing cuts and sores
- 6) Headaches
- 7) Blurred vision
- 8) Trouble concentrating

Low blood sugar level: (Check your blood sugar levels if you feel any of the following symptoms)

- 1) weakness
- 2) Feeling sleepy
- 3) Feeling nauseous
- 4) sweating
- 5) feeling tired
- 6) dizziness
- 7) feeling hungry
- 8) feeling shaky or trembling
- 9) becoming easily irritated, tearful, anxious or moody
- 10) a fast or pounding heartbeat (palpitations)
- 11) confused

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BIOCHEMISTRY

URINE PP (POST PRANDIAL BLOOD GLUCOSE LEVEL)

Test Description	Result	Unit	Reference Range
BSL POST PRANDIAL SUGAR	110	mg/dl	90 - 150

Test Information:

A blood sugar test is a procedure that measures the amount of sugar, or glucose, in your blood. Your doctor may order this test to diagnose diabetes. People with diabetes can also use this test to check their diabetes status.

High blood sugar level: (Check your blood sugar levels if you feel any of the following symptoms)

- 1) Increased thirst
- 2) Frequent peeing
- 3) Fatigue (weak, tired feeling)
- 4) Weight loss
- 5) Slow-healing cuts and sores
- 6) Headaches
- 7) Blurred vision
- 8) Trouble concentrating

Low blood sugar level: (Check your blood sugar levels if you feel any of the following symptoms)

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- 2) Feeling sleepy
- 3) Feeling nauseous
- 4) sweating
- 5) feeling tired
- 6) dizziness
- 7) feeling hungry
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BIOCHEMISTRY

RANDOM (BLOOD GLUCOSE LEVEL)

Test Description	Result	Unit	Reference Range
Blood Sugar Level (Random)	80	mg/dl	70 - 150

Test Information:

A blood sugar test is a procedure that measures the amount of sugar, or glucose, in your blood. Your doctor may order this test to diagnose diabetes. People with diabetes can also use this test to check their diabetes status.

High blood sugar level: (Check your blood sugar levels if you feel any of the following symptoms)

- 1) Increased thirst
- 2) Frequent peeing
- 3) Fatigue (weak, tired feeling)
- 4) Weight loss
- 5) Slow-healing cuts and sores
- 6) Headaches
- 7) Blurred vision
- 8) Trouble concentrating

Low blood sugar level: (Check your blood sugar levels if you feel any of the following symptoms)

- 1) weakness
- 2) Feeling sleepy
- 3) Feeling nauseous
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BIOCHEMISTRY

Blood Sugar Level (Fasting & Post Lunch)

Test Description	Result	Unit	Reference Range
GLUCOSE FASTING	90	mg/dl	70 - 110
GLUCOSE POST PRANDIAL	110	mg/dl	90 - 140

Test Information:

A blood sugar test is a procedure that measures the amount of sugar, or glucose, in your blood. Your doctor may order this test to diagnose diabetes. People with diabetes can also use this test to check their diabetes status.

High blood sugar level: (Check your blood sugar levels if you feel any of the following symptoms)

- 1) Increased thirst
- 2) Frequent peeing
- 3) Fatigue (weak, tired feeling)
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- 6) Headaches
- 7) Blurred vision
- 8) Trouble concentrating

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Blood Sugar Level (Fasting & Post Lunch)

Test Description	Result	Unit	Reference Range
GLUCOSE FASTING	90	mg/dl	70 - 110
GLUCOSE POST PRANDIAL	110	mg/dl	90 - 140

Test Information:

A blood sugar test is a procedure that measures the amount of sugar, or glucose, in your blood. Your doctor may order this test to diagnose diabetes. People with diabetes can also use this test to check their diabetes status.

High blood sugar level: (Check your blood sugar levels if you feel any of the following symptoms)

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BIOCHEMISTRY

RENAL(KIDNEYS) FUNCTION TEST

Test Description	Result	Unit	Reference Range
BLOOD UREA LEVEL (BUL)	20	mg/dl	0 - 45
BLOOD UREA NITROGEN (BUN)	9.35	mg/dl	3.3 - 18.7
CREATININE - SERUM	0.10	mg/dl	0.8 - 1.4
URIC ACID	1.9	mg/dl	2.1 - 7.5
CALCIUM	6.2	mg/dl	8.5 - 11.0

Test Information

Kidneys play several vital roles in maintaining health. One of their most important jobs is to filter waste materials from the blood and expel them from the body as urine. The kidneys also help control the levels of water and various essential minerals in the body. In addition, they're critical to the production of:

- 1) vitamin D
- 2) red blood cells
- 3) hormones that regulate blood pressure

Patient also need kidney function testing done if he/she has other conditions that can harm the kidneys, such as diabetes or high blood pressure.

Symptoms that may indicate a problem with your kidneys include:

- 1) high blood pressure
- 2) blood in the urine
- 3) frequent urges to urinate
- 4) difficulty beginning urination
- 5) painful urination
- 6) swelling of the hands and feet due to a buildup of fluids in the body

A single symptom may not mean something serious. However, when occurring simultaneously, these symptoms suggest that your kidneys aren't working properly. Kidney function tests can help determine the reason.

METHOD :

BLOOD UREA NITROGEN (BUN) - KINETIC UV ASSAY.
CREATININE- CREATININE ENZYMATIC METHOD
URIC ACID - URICASE / PEROXIDASE METHOD
CALCIUM - ARSENAZO III METHOD, END POINT.

END OF REPORT

BIOCHEMISTRY

LIVER FUNCTION TEST (LFT)			
Test Description	Result	Unit	Reference Range
Bilirubin Total	0.9	mg/dL	0.2 - 1.2
Bilirubin Direct	0.2	mg/dL	0.0 - 0.4
Bilirubin Indirect	0.70	mg/dL	0.1 - 1.0
SGOT (AST)	20	U/L	0 - 45
SGPT (ALT)	20	U/L	0 - 45
GAMMA GLUTAMYL TRANSFERASE (G.G.T.)	75	U/L	5 - 85
Alkaline Phosphatase	90	U/L	40 - 130
Protein Total	5.3	g/dL	6.3 - 8.4
Albumin	2.5	g/dL	3.2 - 5.0
Globulin	2.80	g/dL	1.9 - 3.9
A/G Ratio	0.89	Ratio	1.0 - 2.1

Test Information

Liver function tests help determine the health of the liver by measuring the levels of proteins, liver enzymes, or bilirubin in the blood.

Liver tests can help determine is the liver is working correctly. The liver performs a number of vital bodily functions, such as:

- 1) removing contaminants from the blood
- 2) converting nutrients from the foods eaten
- 3) storing minerals and vitamins
- 4) regulating blood clotting
- 5) producing proteins, enzymes, and bile
- 6) making factors that fight infection

Symptoms of a liver disorder include:

- 1) weakness
- 2) fatigue or loss of energy
- 3) jaundice (yellow skin and eyes)
- 4) abdominal pain
- 5) discolored bodily discharge (dark urine or light stools)
- 6) symptoms of the nephritic syndrome (swelling around the eyes, belly, and legs)
- 7) nausea/ vomiting

METHOD :

TOTAL BILIRUBIN - VANADATE OXIDATION

BILIRUBIN -DIRECT - VANADATE OXIDATION

BILIRUBIN (INDIRECT) - DERIVED FROM SERUM TOTAL AND DIRECT BILIRUBIN VALUES

SGOT - IFCC* WITHOUT PYRIDOXAL PHOSPHATE ACTIVATION

SGPT - IFCC* WITHOUT PYRIDOXAL PHOSPHATE ACTIVATION

ALKALINE PHOSPHATASE- MODIFIED IFCC METHOD

PROTEIN- BIURET METHOD

ALBUMIN - ALBUMIN BCG¹METHOD (COLORIMETRIC ASSAY ENDPOINT)

GLOBULIN - DERIVED FROM SERUM ALBUMIN AND PROTEIN VALUES

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ENDOCRINOLOGY

THYROID FUNCTION TEST (TFT)

Test Description	Result	Unit	Reference Range
TOTAL TRIIODOTHYRONINE (T3) Competitive Chemi Luminescent Immuno Assay	110	ng/dl	40 - 181
TOTAL T4 Competitive Chemi Luminescent Immuno Assay	10.4	µg/dL	4.5 - 14.5
THYROID STIMULATING HORMONE (TSH) SANDWICH CHEMI LUMINESCENT IMMUNO ASSAY	2.6	µIU/ml	0.5 - 8.9

Thyroid : The principal function of the thyroid gland is the synthesis and secretion of thyroid hormones, thyroxine (T4) and triiodothyronine (T3), which circulate in the body through the blood stream and aid in overall body metabolism. They are controlled by another hormone called TSH (Thyroid Stimulating Hormone) from pituitary gland. The values helps in evaluating function of the thyroid gland- hypothyrodism or hyperthyrodism.

Hypothyrodism : Hypothyroidism occurs when your body doesn't produce enough thyroid hormones. The most common signs and symptoms of hypothyroidism include:

1) fatigue	9) depression
2) constipation	10) feeling cold
3) dry skin	11) weight gain
4) muscle weakness	12) decreased sweating
5) slowed heart rate	13) elevated blood cholesterol
6) dry, thinning hair	14) impaired memory
7) fertility difficulties or menstrual changes	15) hoarseness of voice
8) muscle stiffness, aches, and tenderness	16) puffy, sensitive face

Hyperthyroidism is a condition in which the thyroid overproduces hormones. The most common signs and symptoms of Hyperthyroidism include:

1) Person may experience a rapid heart rate	8) elevated blood pressure
2) hand tremors	9) more frequent bowel movements
3) weight loss	10) in women irregular menstrual cycles.
4) increased appetite	11) nervousness
5) inability to concentrate	12) weakness
6) difficulty sleeping	13) fine, brittle hair, hair loss
7) irregular heartbeat	14) restlessness

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ENDOCRINOLOGY

ULTRA TSH (THYROID STIMULATING HORMONE)

Test Description	Result	Unit	Reference Range
ULTRA THYROID STIMULATING HORMONE (TSH) FULLY AUTOMATED CHEMILUMINESCENT MICROPARTICLE IMMUNOASSAY	4.6	μIU/mL	0.35 - 5.5

Pregnancy reference ranges for TSH
1st Trimester : 0.10 - 2.50
2nd Trimester : 0.20 - 3.00
3rd Trimester : 0.30 - 3.00

TEST	1 - 3 D	4 - 30 D	31 - 60 D	61 D - 12 M	1 - 5 Y	6 - 10 Y	11 - 14 Y	15 - 18 Y
TSH	0.1-9.2	0.2-8.5	0.2-7.8	0.30-5.9	0.4-4.8	0.5-4.7	0.5-4.6	0.6-4.5
T3	41.7-272.1	48.2-272.1	54.7-272.1	76.8-272.1	89.2-246.7	87.2-218.1	86.6-199.8	85.3-188.8
T4	4.9-15.8	5-15.3	5.2-14.8	5.7-13.3	5.7-11.7	5.4-10.7	5.2-10	5.1-9.6
FT3	1.5-5.3	1.6-5.2	1.6-5.1	1.8-4.8	2-4.5	2.1-4.4	2.3-4.4	2.3-4.3
FT4	0.84-2.08	0.85-1.98	0.85-1.89	0.89-1.62	0.89-1.48	0.85-1.46	0.84-1.45	0.84-1.45

Thyroid : The principal function of the thyroid gland is the synthesis and secretion of thyroid hormones, thyroxine (T4) and triiodothyronine (T3), which circulate in the body through the bloodstream and aid in overall body metabolism. They are controlled by another hormone called TSH (Thyroid Stimulating Hormone) from the pituitary gland. The values help in evaluating the function of the thyroid gland- hypothyroidism or hyperthyroidism.

Hypothyroidism occurs when your body doesn't produce enough thyroid hormones. The most common signs and symptoms of hypothyroidism include

1) fatigue	9) depression
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5) slowed heart rate	13) elevated blood cholesterol
6) dry, thinning hair	14) impaired memory
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8) muscle stiffness, aches, and tenderness	16) puffy, sensitive face

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|---|--|
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| 3) weight loss | 10) in women irregular menstrual cycles. |
| 4) increased appetite | 11) nervousness |
| 5) inability to concentrate | 12) weakness |
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ENDOCRINOLOGY

ULTRA TSH (THYROID STIMULATING HORMONE)

Test Description	Result	Unit	Reference Range
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CLINICAL PATHOLOGY			
URINE ROUTINE			
Test Description	Result	Unit	Reference Range
PHYSICAL EXAMINATION			
Quantity	10	ml	
Colour	Pale Yellow		Pale yellow/Yellow
Appearance	Clear		Clear
Specific Gravity	1.010		1.005-1.030
pH	Acidic		Acidic
Deposit	Absent		Absent
CHEMICAL EXAMINATION			
Protein	Absent		Absent
Sugar	Absent		Absent
Ketones	Absent		Absent
Bile Salt	Absent		Absent
Bile Pigment	Absent		Absent
Urobilinogen	Normal		Normal
MICROSCOPIC EXAMINATION			
Pus Cell	2-4		Upto 5
Epithelial Cells	1-2		Upto 5
Red Blood Cells	Absent		Absent
Casts	Absent		Absent
Crystals	Absent		Absent
Bacteria	Absent		Absent

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IMMUNOASSAY

VITAMIN B12

Test Description	Result	Unit	Reference Range
Vitamin B12 Method: CLIA	400	pg/mL	120 - 807

General information:

Like all B vitamins, vitamin B12 is a water-soluble vitamin. This means it dissolves in water and travels through the bloodstream. The human body does not store cobalamin. Any excess or unwanted amounts are excreted through urine

Uses:

- 1) Vitamin B12 plays a vital role in the normal functioning of the brain and the nervous system and in the formation of red blood cells. It also helps to regulate and synthesize DNA.
- 2) It is needed in the metabolism of every cell in the body, and it plays a part in the synthesis of fatty acids and energy production. By helping the human body to absorb folic acid, it facilitates the release of energy.
- 3) Heart and entire cardiovascular system needs B12. One of its jobs is to remove a dangerous protein called homocysteine from the blood. If homocysteine is allowed to roam through blood, it damages your arteries leading to inflammation and heart disease.

Deficiency

- 1) Vitamin B12 deficiency can result in irreversible and potentially severe damage, especially to the human nervous system and brain. It can also cause anemia. Even at slightly lower-than-normal B12 levels, a person can experience depression, confusion memory problems and tiredness.
- 2) More serious symptoms include neurological changes, such as numbness and tingling in the hands and feet. Some people may have difficulty maintaining balance.
- 3) The human body produces millions of red blood cells every minute, but without vitamin B12, cells cannot multiply properly. The production of red blood cells goes down if a person's vitamin B12 levels are too low. As the red blood cell count drops, anemia results.

Method: Fully Automated Bidirectionally Interfaced ChemiLuminescent Immuno Assay

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IMMUNOASSAY

VITAMIN D3			
Test Description	Result	Unit	Reference Range
VITAMIN D3	80	ng/ml	DEFICIENCY : <20 ng/ml INSUFFICIENCY : 20-30 ng/ml SUFFICIENCY : 30-100 ng/ml TOXICITY : >100 ng/ml

Vitamin D is essential for the formation and maintenance of strong, healthy bones.

Vitamin D is a fat-soluble vitamin that helps your body absorb calcium and phosphorus. Having the right amount of vitamin D, calcium, and phosphorus is important for building and keeping strong bones. Vitamin D is used to treat and prevent bone disorders (such as rickets, osteomalacia). Vitamin D is made by the body when skin is exposed to sunlight. Sunscreen, protective clothing, limited exposure to sunlight, dark skin, and age may prevent getting enough vitamin D from the sun.

Vitamin D with calcium is used to treat or prevent bone loss (osteoporosis). Vitamin D is also used with other medications to treat low levels of calcium or phosphate caused by certain disorders (such as hypoparathyroidism, pseudohypoparathyroidism, familial hypophosphatemia). It may be used in kidney disease to keep calcium levels normal and allow normal bone growth. Vitamin D drops (or other supplements) are given to breast-fed infants because breast milk usually has low levels of vitamin D.

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BIOCHEMISTRY

IRON DEFICIENCY PROFILE

Test Description	Result	Unit	Reference Range
IRON	75	µg/dl	65 - 175
Method : FERROZINE METHOD WITHOUT DEPROTEINIZATION			

A serum iron test measures how much iron is in your serum. Having too much iron or not enough iron can cause serious health problems. This test will help doctor give a more accurate diagnosis.

Early symptoms of iron deficiency (anemia) include:

- 1) chronic fatigue
- 2) dizziness
- 3) headaches
- 4) muscle weakness

May develop other symptoms as condition worsens. These may include:

- 1) difficulty concentrating
- 2) tongue and mouth sores
- 3) pica (the compulsion to eat non-food items, such as paper)
- 4) misshapen nails

Symptoms of iron overload (when body produces too much iron) include:

- 1) pain in your abdomen and joints
- 2) bronzing or darkening of skin
- 3) fatigue
- 4) heart problems
- 5) lack of energy
- 6) lack of sex drive
- 7) weight loss
- 8) muscle weakness

What Is a Total Iron Binding Capacity (TIBC)

Iron is an important mineral found in all cells of the body. It's needed to produce hemoglobin that carries oxygen through the bloodstream. In the blood, a protein called transferrin, which is made by the liver, carries iron. A TIBC test measures how much iron is carried in your bloodstream. Low TIBC levels usually indicate high levels of iron in the blood. High iron binding capacity levels typically indicate low levels of iron in the blood.

Transferrin saturation , measured as a percentage. It is the value of serum iron divided by the total iron-binding capacity of the transferrin that is available to bind iron, this value tells how much serum iron is bound.

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BIOCHEMISTRY

ELECTROLYTES

Test Description	Result	Unit	Reference Range
Sodium	140	mEq/L	135 - 155
Potassium	4.5	mEq/L	3.5 - 5.5
Chloride	100	mEq/L	94 - 110

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BIOCHEMISTRY

RA FACTOR (RHEUMATOID ARTHRITIS FACTOR) Quantitative

Test Description	Result	Unit	Reference Range
Sample Type :	Serum		
Result:	Negative		
Titre :	12	IU/ml	-
Interpretation :	More than 20 IU/ML is positive		
Method :	Turbidimetry		

Test Information:

Rheumatoid factor is an antibody directed against the Fc portion of the IgG molecule. Polyreactive RF has binding specificity for substances other than IgG like nuclear components. This polyreactive RF is usually of the IgM class with low affinity. RF is not specific only for Rheumatoid arthritis, but it is often seen in cases of chronic infection and other systemic inflammatory conditions. Healthy individuals > 65 years of age may also show positive RF results. In addition to the common IgM RF, both IgA RF & IgG RF have been detected. IgA RF has been related to the more severe form of the disease with erosions.

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BIOCHEMISTRY			
C REACTIVE PROTEIN (CRP)			
Test Description	Result	Unit	Reference Range
Sample Type:	SERUM		
Result:	Negative		
Titre	0.2	MG/L	
Interpretation	UP TO 6 MG/L is Negative		
Method	Turbidimetry		

Test Information

- 1) C-reactive protein (CRP) is a protein found in the blood, the levels of which rise in response to inflammation (an acute-phase protein).
- 2) Its physiological role is to bind to phosphocholine expressed on the surface of dead or dying cells (and some types of bacteria) in order to activate the complement system via c1q. CRP is synthesized by the liver in response to factors released by fat cells (adipocytes).
- 3)It is a member of the pentraxin family of proteins. It is not related to C-peptide or protein C. CRP is used mainly as a marker of inflammation. Apart from liver failure, there are few known factors that interfere with CRP production.
- 4Measuring and charting CRP values can prove useful in determining disease progress or the effectiveness of treatments.
- 5)CRP is therefore a test of value in medicine, reflecting the presence and intensity of inflammation,although an elevation in C-reactive protein is not the telltale diagnostic sign of any one condition.

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SEROLOGY

HIV 1 & 2 Antibody			
Test Description	Result	Unit	Reference Range
Sample type :	Serum		
Antibodies to H.I.V. - I	Not detected		
Antibodies to H.I.V. - II	Not detected		
Impression :	Negative		
Method:	HIV (TRI-DOT)		

Test information

1. The above test is screening test for detection of HIV 1 and HIV 2 in human serum or plasma using recombinant proteins immobilized on an immunofiltration membrane
2. Their antibodies have a cross reactivity of 30 - 70 % when tested using recombinant proteins.
3. Appearance of dots of HIV I & HIV 2 antibodies on the test device does not necessarily imply co-infection from HIV-1 and HIV-2.
4. This is only screening test All positive detected sample shall be reconfirmed by using WESTERN BLOT techniques
5. A negative test result does not exclude the possibility of infection or exposure to HIV.

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SEROLOGY

AUSTRALIA ANTIGEN (HBsAg)

Test Description	Result	Unit	Reference Range
Sample Type:	Serum		
Result:	Negative		
Method:	Immunochromatography		

Test Information

1)Hepatitis B surface antigen (HBsAg) is an important viral envelope protein,which appears shortly after infection and is a key serological marker for detection and diagnosis of HBV.Clearance during treatment shows recovery and development of neutralizing antibodies (anti-HBs) occurs in 90% of the patients.due to the introduction of hepatitis B vaccination programs,the serological detection of anti-HBs has become important method for monitoring of recipients upon vaccination with synthetic and natural HbsAg.

2)The absence of anti-HBs indicates susceptibility to HBv infection.For this screening for anti-HBs in high risk populations is recommended for identifying individuals who may benefit from vaccination.

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IMMUNOASSAY

HEPATITIS C VIRUS (HCV) IgM Ab

Test Description	Result	Unit	Reference Range
Sample Type :	Serum		
Method :	Immunoassay		
Result :	Negative		

NOTE : 1)Hepatitis C virus (HCV) is now know to be the major cause of parenterally transmitted non-A, non-B hepatitis.

2) Until the virus was characterized, diagnosis was made by exclusion of all other known causes of hepatitis.

3) Antibody to HCV is found in over 80% of patients with well documented non-A non-B hepatitis. The Worldwide prevalence of HCV is 0.2 to 2% in blood donors and up to 80% in intravenous- drug users. Seroepidemiologic studies show that the seroprevalence of HCV infection in India varies between 0.3% to 11.3%. In a large percentage of HCV cases transmission by transfusion and other parenteral means such as sharing of neediest, occupational exposure to blood and hemodialysis.

4)However in case of half of HCV infections, the route of transmission is unknown. HCV establishes a chronic infection in 50 to 80% of cases.

5)Chronic infection is often asymptomatic even in the presence of liver damage discernible on biopsy. Chronic HCV is characterized by fluctuating alanine aminotransferase (ALT or SGPT) levels and recognizable changes in liver histology. Chronic infection can lead to cirrhosis and hepatocellular carcinoma

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HEMATOLOGY

RAPID MALARIA TEST

Test Description	Result	Unit	Reference Range
RAPID MALARIA TEST	Negative		

Test Information

- 1)Four species of the Plasmodium parasites are responsible for malaria infections in human viz. P. falciparum, P.vivax, P.ovale and P.malariae.
- 2)Of these, P. falciparum and P.vivax are the most prevalent. Early detection and differentiation of malaria is of utmost importance due to incidence of cerebral malaria and drug resistance associated with falciparum malaria and due to the morbidity associated with the other malarial forms.
- 3) As the course of treatment is dependent on the species, differentiation between P. falciparum and P.vivax is of utmost importance for better patient management and speedy recovery.
- 4)In Core Malaria Pan/Pv/Pf the detection system for P. falciparum malaria is based on the detection of P. falciparum specific histidine rich protein -2 (Pf HRP-2) which is a water soluble protein that is released from parasitised erythrocytes of infected individuals.
- 5)The detection system of P.vivax is based on the presence of P.vivax specific pLDH. Further the detection of other malarial infections such as P.ovale and P.malariae is achieved through the pan malaria specific pLDH.
- 6) Since pLDH is a product of viable parasites, the pan band may also be used to monitor course of effective antimalarial therapy. Core Malaria Pan/Pv/Pf detects the presence of P. falciparum specific Pf.HRP-2, P.vivax specific pLDH and pan specific pLDH in whole blood specimen and is a sensitive and specific test for the detection of all malaria species, differentiation for P.falciparum and P.vivax and monitoring successful antimalarial therapy.

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HEMATOLOGY

PBS MALARIAL PARASITE

Test Description	Result	Unit	Reference Range
Malaria Parasite	Not Detected		
Note	Repeat PBS/Rapid maleria test s.o.s.		

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IMMUNOASSAY

Dengue IgG & IgM Antibody (card Test)

Test Description	Result	Unit	Reference Range
Dengue IgM	Negative		
Dengue IgG	Negative		

Test Information

- 1.The analysis of single serum sample should not be used as the sole criterion for diagnosis.
2. In early infection and some secondary infections, detectable levels of IgM antibodies may be low. Some patients may not produce detectable levels of antibody within the first seven to ten day after infection. Where symptoms persist, patients should be re-tested 3 - 4 days after the first specimen.
3. Heterophilic antibodies are a well-recognized cause of interference in immunoassays. These antibodies may cross-react with reagent antibodies and generate false positive signal.
4. The final diagnosis should be based on test results in conjunction with other clinical and laboratory findings.
5. Dengue immunochromatography is a screening assay. The diagnosis of acute dengue infection should be confirmed by IgM ELISA or by other tests like Real Time PCR.

TECHNIQUES & KITS USED Bioline : Dengue IgG/IgM (Standard Diagnostic)

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IMMUNOASSAY

Dengue Ns1 Antigen (card Test)

Test Description	Result	Unit	Reference Range
Dengue NS1 Ag	Negative		

Test information

Primary dengue virus infection is characterized by elevations in specific NS1 antigen levels 0 to 9 days after the onset of symptoms; this generally persists upto 15 days.

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CLINICAL PATHOLOGY

WIDAL TEST AGGLUTINATION

Test Description	Result	Unit	Reference Range
S. Typhi 'O'	No Agglutination		No Agglutination
S. Typhi 'H'	No Agglutination		No Agglutination
S.Paratyphi A(H)	No Agglutination		No Agglutination
S.Paratyphi B(H)	No Agglutination		No Agglutination
Result :	Test is Negative		

Test Information

Agglutination titer greater than 1:80 is considered significant.
The past history of enteric fever or inoculation of vaccine can interfere with the result.

Clinical significance:

1. The organism Salmonella typhi is responsible for causing enteric fever/typhoid fever, which is characterized by high consistent fever, loss of appetite, transitory bacteremia, round or oval-shaped ulcers on the smooth peritoneal surface of Peyer's patches, etc.
2. The organism possesses an 'O' antigen on the cell wall and 'H' antigen on its flagella, against which the host body produces immunospecific antibodies.
3. Paratyphoid fever caused by Salmonella paratyphi A or B is characterized by a milder course of the disease.
4. These organisms also possess somatic 'O' and flagellar antigen termed as A(H) and B (H) respectively

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IMMUNOASSAY

Widal IgG IgM

Test Description	Result	Unit	Reference Range
Widal IgG	Negative		Negative
Widal IgM	Negative		Negative

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SEROLOGY

VDRL

Test Description	Result	Unit	Reference Range
Sample Type	Serum		
Result	Negative		

False positive are common in rheumatic disease and auto-immune disorders.
False negative can occur in very early stages of the in latent stages.
TPHA test is recommended for conclude.

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SEROLOGY

ANTI HEPATITIS C VIRUS TOTAL

Test Description	Result	Unit	Reference Range
ANTI HEPATITIS C VIRUS (HCV) - TOTAL Method: CMIA	0.5	S/CO	Reactive : >= 1.0 Non Reactive : <1.0

Clinical Significance:
Hepatitis C (HCV) is a virus that causes an infection of the liver that is characterized by liver inflammation and damage. An HCV antibody test is typically reported as "positive" or "negative."If no virus is present or if the amount of virus is too low to detect, the result is often reported as "negative" or "not detected."In general, if the HCV antibody test is positive, then the individual tested is infected or has likely been infected at some time with hepatitis C.

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CLINICAL PATHOLOGY

BLOOD GROUP			
Test Description	Result	Unit	Reference Range
Sample Type:	Whole Blood Edta		
Blood Group :	AB Rh Positive		
Method:	Monoclonal blood grouping (Agglutination test) by slide method		
Kit :	Span diagnostics.		

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BIOCHEMISTRY

CREATININE

Test Description	Result	Unit	Reference Range
Serum Creatinine	1.0	mg/dl	0.8 - 1.4

Test information

A creatinine blood test measures the level of creatinine in the blood. Creatinine levels in the blood can provide doctor with information about how well kidneys are working.

Why is a creatinine blood test done?

Doctor may order a creatinine blood test to assess creatinine levels if show signs of kidney disease. These symptoms include:

- fatigue and trouble sleeping
- a loss of appetite
- swelling in the face, wrists, ankles, or abdomen
- lower back pain near the kidneys
- changes in urine output and frequency
- high blood pressure- nausea- vomiting

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BIOCHEMISTRY

CALCIUM

Test Description	Result	Unit	Reference Range
Calcium	9.6	mg/dl	8.5 - 11.0

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BIOCHEMISTRY

SGPT

Test Description	Result	Unit	Reference Range
SGPT (ALT)	36	U/L	0 - 45

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BIOCHEMISTRY

SGOT

Test Description	Result	Unit	Reference Range
SGOT (AST)	36	U/L	5 - 45

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BIOCHEMISTRY

BILIRUBIN			
Test Description	Result	Unit	Reference Range
Bilirubin Total (Serum)	0.9	mg/dL	0.2 - 1.2
Bilirubin Direct	0.2	mg/dL	0.0 - 0.3
Bilirubin Indirect	0.70	mg/dL	0.2 - 0.9

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BIOCHEMISTRY

URIC ACID

Test Description	Result	Unit	Reference Range
Uric Acid	5.6	mg/dl	2.1 - 7.5

Test Information

The test can help determine how well your body produces and removes uric acid. Most uric acid is dissolved in the blood, filtered through the kidneys, and expelled in the urine. Sometimes, the body produces too much uric acid or doesn't filter out enough of it. Hyperuricemia is the name of the disorder that occurs when you have too much uric acid in your body.

High levels of uric acid are associated with a condition called gout. Gout is a form of arthritis that causes swelling of the joints, especially in the feet and big toes. Another cause of hyperuricemia is increased cell death, due to cancer or cancer treatments. This can lead to an accumulation of uric acid in the body.

Most commonly, the test is used to:

- 1)diagnose and monitor people with gout
- 2)monitor people who are undergoing chemotherapy or radiation treatment
- 3)check kidney function after an injury
- 4)find the cause of kidney stones 5)diagnose kidney disorders

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BIOCHEMISTRY

ALKALINE PHOSPHATASE

Test Description	Result	Unit	Reference Range
Alkaline Phosphatase (Serum)	96	U/L	46 - 116

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BIOCHEMISTRY

CHOLESTEROL

Test Description	Result	Unit	Reference Range
Cholesterol Total (Serum)	165	mg/dl	Low : < 125 Desirable : < 200 Borderline high : 201 - 240 High : > 240

END OF REPORT

HEMATOLOGY

GLYCOSYLATED HAEMOGLOBIN (GHB / HBA1c)

Test Description	Result	Unit	Reference Range
HbA1c	2.4	%	Below 6.0% - Normal Value
Method: HPLC			6.0% - 7.0% - Good Control
			7.0% - 8.0% - Fair Control
			8.0% - 10% - Unsatisfactory Control
			Above 10% - Poor Control

Technology : H.P.L.C
Method : Fully Automated H.P.L.C. using Biorad Variant II Turbo

AVERAGE BLOOD GLUCOSE (ABG)	22.18	mg/dl	90 - 120 mg/dl : Excellent Control
Derived from HBA1c values			121 - 150 mg/dl : Good Control
			151 - 180 mg/dl : Average Control
			181 - 210 mg/dl : Action Suggested
			>211 mg/dl : Panic Value
			(Note: Average Blood Glucose value is calculated from HBA1c value and it indicates Average Blood Sugar level over past three months.)

Test Information

People with diabetes used to measure their sugar level as a fasting, Postprandial or Random from their blood sample. These tests are accurate, but only at the moment. As an overall measurement of blood sugar control, they're very limited. This is because blood sugar can vary wildly depending on the time of a day, activity levels.

HbA1C became an important tool in controlling diabetes.

HbA1C became an important tool in controlling diabetes.HbA1C tests measure average blood glucose over the past two to three months. So even if you have a highfasting blood sugar, your overall blood sugars may be normal, or vice versa.A normal fasting blood sugar may not eliminate the possibility of type 2 diabetes. This is why HbA1C testsare now being used for diagnosis and screening of prediabetes. Because it doesn't require fasting, the testcan be given as part of an overall blood screening.

High HbA1c levels have been linked to increase a risk of the following conditions:

- 1) cardiovascular diseases, such as a stroke and the heart attack.
- 2) a kidney disease
- 3) nerve damage
- 4) eye damage that may result in blindness
- 5) foot problems due to nerve damage that can cause numbness, tingling, and lack of sensations in the feet
- 6) slower wound healing that may lead to infection.

For those who have had prediabetes or diabetes for a long time, HbA1c results that are moving upward is a sign that you need to start the medication or change what you're already taking.

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HEMATOLOGY

GLYCOSYLATED HAEMOGLOBIN (GHB / HBA1c)

Test Description	Result	Unit	Reference Range
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PT INR			
Test Description	Result	Unit	Reference Range
Sample Type	Sodium Citrate		
Patients Prothrombin Time	16	seconds	12 - 16
Control Prothrombin Time	14	Seconds	-
ISI VALUE	1.1	-	-
INR	1.16	-	Standard Therapy: INR = 2 to 3 High Dose Therapy: INR = 3 to 4.5

Note ; T PT is the time taken by the liquid portion [plasma] of blood to form the clot. It is the measure of extrinsic pathway of coagulation. PT is prolonged in deficiency of clotting factors Fibrinogen I, Prothrombin II, Factor V, Factor VII and Factor X. It is also prolonged in Vit. K deficiency, liver damage, DIC, Patients on Anticoagulant Therapy and Congenital Afibrinogenemia

END OF REPORT